

Planar Reflection

Simulating Mirror Surfaces in a 3D Rendering Pipeline

Use cases :

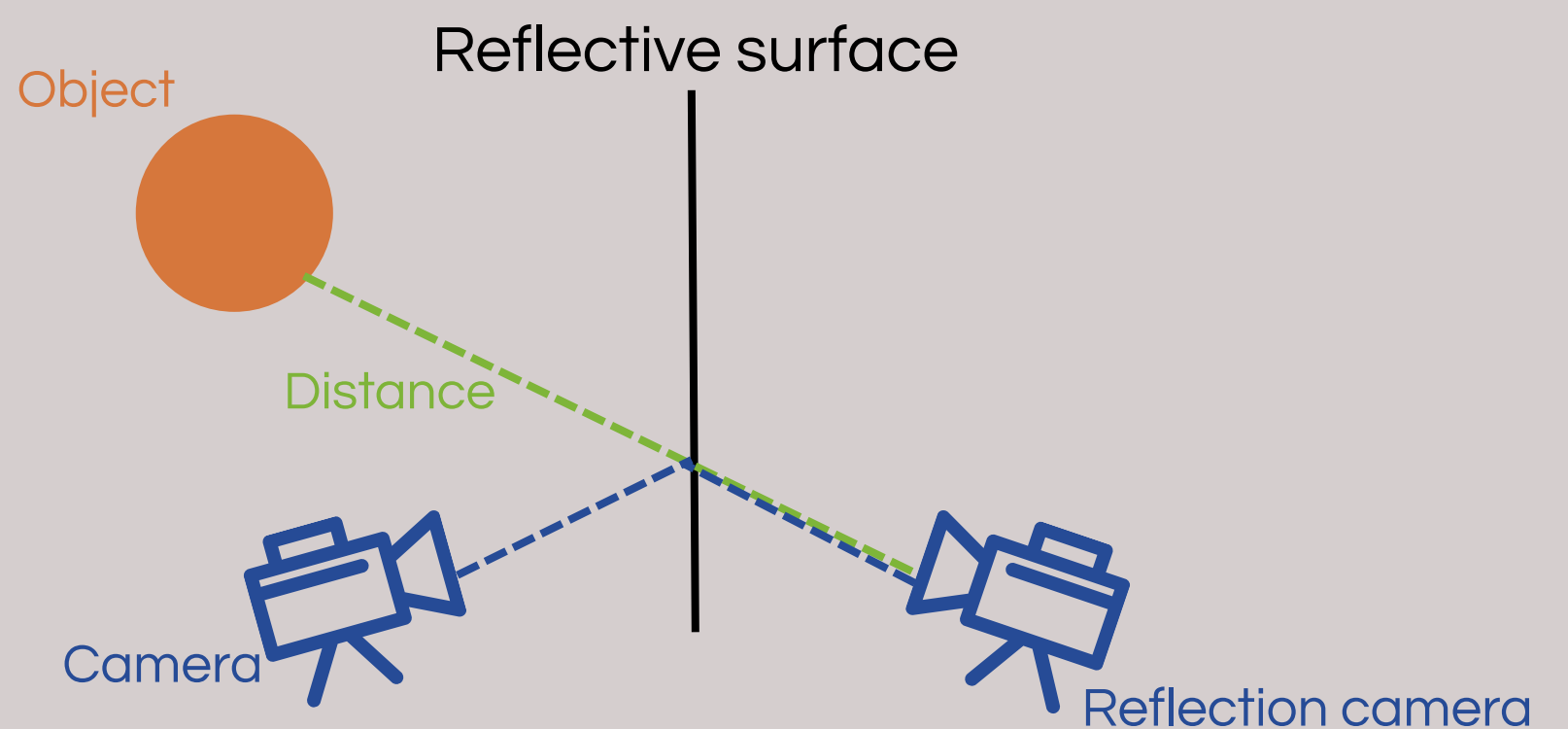
- Reflective floors, puddles, indoor tiling
- Material simulation (marble, glass, polished metal)
- Mirrors, calm water surfaces, display windows

Method:

- Virtual camera
 - Symmetric to the real camera with respect to the mirror plane
 - Full render of the scene from the virtual camera
- Clipping plane
 - Clips everything on the wrong side of the mirror plane
- Screen-space sampling
 - Color read from the virtual buffer at coordinates (x, y)
 - Approximation — slight offset on highly inclined surfaces
- Physical effects
 - Fresnel — stronger reflection at grazing angles
 - Attenuation — reflection weakens with distance
 - Roughness — blur on the reflection buffer

What is a planar reflection?

Planar reflection is a 3D rendering technique that simulates the reflection of a scene on a flat reflective surface. It consists of rendering the scene a second time from a virtual camera, symmetric to the real camera with respect to the mirror plane, then projecting this render onto the reflective surface.



Base rendering vs Planar Reflection

Base rendering



Planar reflection



- Reflection appears
- opposite face visible

Results:

The following demonstrate the visual improvement brought by planar reflection over the base rendering pipeline, as well as the impact of each physical effect implemented.

Motivation:

- Adds realistic reflections absent from the base pipeline
- Reflects real-world phenomena everyone can relate to (puddles, mirrors, polished floors)
- Easily extended with physical effects (Fresnel, attenuation, roughness)
- Immediate visual impact — the improvement over the base pipeline is obvious at a glance

Limitations:

- Only works on flat surfaces — no curved objects
- One full extra render per mirror — high performance cost
- Screen-space sampling is an approximation
- Reflection only visible where the mirror quad is rasterized

Improved version:

- Oblique projection — eliminates clipping artifacts near the mirror plane
- Precise UV sampling — projects each fragment through the virtual camera matrix for exact sampling
- Screen-space Reflection (SSR) — works on any surface, no virtual camera needed
- Environment map — reflections on non-flat objects

Références:

- Lengyel, E. Oblique View Frustum Depth Projection and Clipping. Journal of Game Development, 2005.
- Unity Technologies. Screen Space Reflection. Unity Manual.
- Epic Games. Screen Space Reflections in Unreal Engine. Epic Games Documentation.
- de Vries, J. LearnOpenGL: Cubemaps. LearnOpenGL.

Fresnel effect

Fresnel: face-on (weak-reflection)



- In front of the plane surface (object behind camera)
- See the color of the surface
- Low reflection

Fresnel: grazing angle (strong reflection)



- At a grazing edge
- The surface color contribution is lower
- High reflection

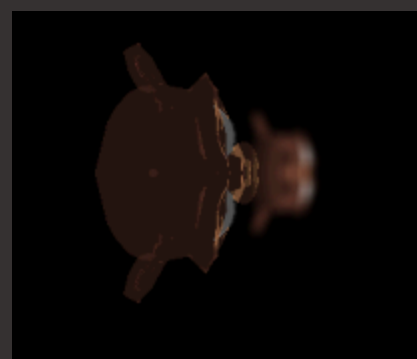
Surface roughness

- Sharp reflection
- Smoother surface
- Reflected light remains coherent

No roughness



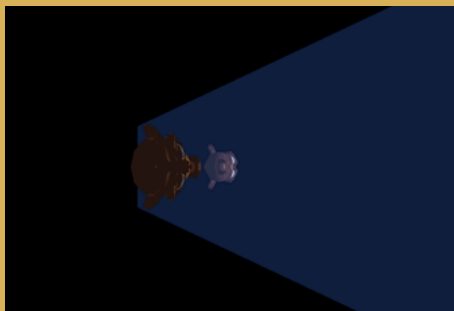
Roughness



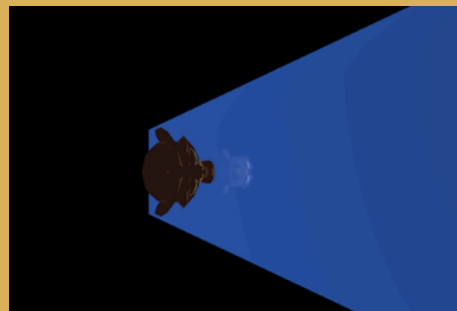
- Blurred reflection
- Rougher surface
- Reflected light is scattered

Attenuation

Without attenuation



With attenuation



- Weak reflection
- Attenuation enabled
- Distance impacts

- Strong reflection
- Attenuation disabled